
ASSESSING PUBLIC TRANSPORT INFRASTRUCTURE: THE ROLE OF EMPLOYMENT MATCHING IN SPATIAL ACCESSIBILITY MEASURES

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Abstract

The definition of accessibility emphasises the role of the attractiveness of opportunities at potential destinations that residents consider valuable. This study revises the common assumption in empirical studies that residents are equally attracted to all types of employment and examines its implications for public transport evaluation. Additionally, the role of the modifiable areal unit problem is also explored in this relationship.

Keywords Accessibility · Employment matching · Public transport infrastructure · Modifiable areal unit problem

1 Introduction

Location-based accessibility measures are valuable for assessing the benefits enabled by transport infrastructure, among others, those in the property market and social equity. At its core, accessibility concerns the ease of travel to opportunities that people consider valuable. This study revises the implications of the common assumption in empirical studies that residents are equally attracted to all types of employment. We focus on employment because it is one of the most frequent purpose for travelling. Additionally, the role of the modifiable areal unit problem (MAUP) is examined in this context. These considerations can provide valuable

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guidance for measuring accessibility in environmental valuation studies or for equity assessments of public transport infrastructure, for example. Thus, the following questions are examined:

1. What are the key spatial differences between generic and matching accessibility measures and what is the role of the MAUP?
2. How does the evaluation of PT infrastructure differs when accounting for matching employment opportunities in accessibility measures?
3. What are the social equity implications of the above?

This paper contributes to literature by examining the implications of relaxing the assumption that all employment opportunities are equally attractive to all the population in accessibility measures drawing on an empirical case for a ten-year period. It also investigates how this specifications interact with the MAUP. Additionally, it presents findings for GMC, an understudied mega-city, by identifying and digitally reconstructing in a common format (general transit feed specification, GTFS) a series of temporal stages of the MPTN in GMC for the same ten-year period from 2009 to 2019.

2 Literature review

Levinson and Wu (2020) assert that accessibility measures “the ease of reaching valued destinations” (p. 130). This notion combines two basic elements, namely the ease of travel and the spatial dispersion of opportunities reflected in land use patterns, including services and other people, for example. Furthermore, this definition limits opportunities to those which are deemed valuable. Thus, the *for whom* question becomes relevant. In line with this discussion, Thériault, Voisin, and Des Rosiers (2013) provides a more explicit definition of accessibility, regarding it as “the ability that persons living in that area can, within an acceptable travel time, reach a satisfactory diversity of activity locations that are important to them” (p. 234).

The operationalisation of accessibility has taken various forms in the literature. The most common are infrastructure-based, person-based, place-based, and utility-based measures (Geurs and van Wee 2004). Place-based measures are frequently used due to their consistency with conceptual definitions, flexibility, and ease of interpretation, enhancing their impact on transport policy. These measures characterise space based on mobility constraints or enablers, combined with land use patterns, by dividing regions into spatial units or zones. This review focuses on location-based measures due to their relevance in policy discussions and their ability to account for people’s abilities and preferences (Páez, Scott, and Morency 2012).

2.1 Accessibility measures and empirical application

Location-based measures consist of the sum of the product of two functions, namely $g(\cdot)$ and $f(\cdot)$, as shown in Equation 1 (Páez, Scott, and Morency 2012):

$$A_{ik}^p = \sum_j g(W_{jk})f(c_{ij}^p). \quad (1)$$

Here, A denotes the accessibility to opportunities of type k from location i , from the perspective of person type p . The measure reflects the attractiveness of opportunities W of type k at destination j . The term c_{ij} refers to a function of travel cost between i and j , as experienced by person p .

Equation 1 is often used in literature to measure accessibility to employment as following:

$$A_i = \sum_{j=1}^N E_j f(c_{ij}) \quad (2)$$

The attractiveness of a destination W is expressed by E , usually as the number of employment opportunities at j , i.e. E_j . Travel cost in $f(c_{ij})$ is often represented by travel time.

In Equation 2, households and opportunities are implicitly assumed to be homogeneous, i.e., all types of employment are equally attractive for all types of residents (Houston 2005; Shen 1998). However, contributions from theory highlight the characteristics of the demand (residents at origins) and supply at potential destinations (Geurs and van Wee 2004; Levinson and Wu 2020). Specifically, Levinson and Wu state that “different job categories represent different economic activities, and different types of jobs generally require different skills and education levels” (2020, 147). This distinction is supported by the Human Capital Theory from the view of labour market studies (Schultz 1961; Becker 1994).

Location-based measures can consider some of these distinctive elements. For instance, Equation 1 refers to the following components: (1) individual’s profiles; (2) how different profiles perceive or experience the space, and; (3) constraints that may prevent segments of the population to access some type of activities. Some of these aspects have been found to be relevant in empirical studies. For instance, gender in Osland (2010) or age and gender in Thériault, Voisin, and Des Rosiers (2013) in the study of the property prices.

The operationalization of measures accounting for employment preferences or restrictions can vary. In empirical studies, these characteristics can be expressed as type of person m at j and type of employment m at k (Pereira 2019; Shen 1998; Pan, Jin, and Liu 2020). Accordingly, only employment opportunities E matching the profile of residents between j and k are considered in the accessibility measure. This reflects the notion that residents would find attractive only employment opportunities that match their skills and education level.

In Pereira (2019), an accessibility measure recognizing such characteristics is implemented as a *proxy* match by distinguishing the upper and lower halves of residents’ income and education requirements in employment opportunities. Specifically, the lower half is matched only with employment opportunities requiring secondary or primary education, while the upper half considers those requiring higher or secondary education. Although

the aim of this study is the identification of potential inequalities in transport policy and not the explicit discussion of this measure, the authors report no substantial difference in their conclusions when disregarding the match in accessibility measures.

Levinson and Wu (2020) suggest to weight employment in each category with the number of eligible workers within each corresponding segment. This perspective thus considers two aspects discussed in employment mismatch literature, competition for employment opportunities and labour segmentation. Bian (2020) refers to these aspects as a numerical (objective) imbalance and an expectation imbalance (subjective), respectively.

An empirical application which explores both numerical and expectation imbalance can be found in Pan, Jin, and Liu (2020). To compute accessibility measures in Houston, the study categorises employment opportunities by the number of white- and blue-collar jobs at potential destinations. On the residents' side, it considers unemployed individuals as job seekers, categorising them by education level: *associate's degree or above* and *high school degree or lower*. White-collar jobs are relevant only to job seekers with an associate's degree or higher, while blue-collar jobs are relevant to the less educated group. The measures also account for different transport modes, i.e. automobile, drive to transit, and walk to transit. Results show that accessibility is concentrated in core areas, declining slowly for drivers and much more sharply for transit users. The study concludes that automobile are even more essential for low-educated groups since blue-collar jobs are more dispersed across the city than white-collar jobs. The study concludes low-educated groups are more dependent of car since blue-collar jobs are more dispersed across the city than white-collar jobs.

The study in Pan, Jin, and Liu (2020) follows a 'descriptive' rather than a 'prescriptive' approach (Páez, Scott, and Morency 2012). This limits its use in transport policy evaluation, as it characterises multiple residents within the same location rather than providing a single measure for a determined zone. Additionally, it does not address the differences or implications of using education and employment matching compared to accessibility measures that disregard these distinctions.

2.1.1 The modifiable areal problem unit in accessibility measures

Another aspect of location-based measures is that these tend to characterise the space in discrete units. Thus, these often suffer from the MAUP. The MAUP literature distinguishes two main types of issues, namely *scale* effects and *zoning* effects (Wong 2009a). Scale effects refer to the variability in spatial analysis results depending on the spatial resolution used. Zoning effects, on the other hand, arise from the spatial configuration of areas while maintaining the same number of zones in an area of study. Unless location-based accessibility is calculated at the individual level, it will likely be affected by these issues. Therefore, this is a main consideration for most location-based applications.

Considerable effort has been made to understand the effects and structural patterns of the MAUP. The clearest effects are observed in simple univariate statistics. It has been noted that while mean values remain

fairly constant across scales, variance tends to decline as spatial resolution decreases, with a narrower range at lower resolutions (Dark and Bram 2007; Wong 2009b). However, this structure tends to dissipate in multivariate regression analyses (Fotheringham and Wong 1991). Zoning effects, on the other hand, are less predictable and are influenced by factors such as spatial autocorrelation and aggregation methods, e.g. mean, sum, median (Wong 2009b). Zoning effects are minimal in cases of random spatial patterns, but results are highly sensitive to zoning when strong positive spatial autocorrelation exists (Lee et al. 2019).

Accessibility studies often use a single unit of analysis. Thus, literature is limited in the role of MAUP in location-accessibility (Stepniak and Rosik 2015). An empirical study in Poland examined three gravity-type accessibility measures at different spatial scales. The findings showed that larger units, such as municipalities, produced smoother estimates compared to disaggregated grid-based inputs, consistent with MAUP literature (Wong 2009b). Additionally, differences in self-potential estimates were attributed to the gravity measures' components (e.g., transport and land use) rather than the estimation process itself. Further work has suggested that errors arise from aggregating travel times within zones (intra-zone) and between zones (inter-zone) (Stepniak and Jacobs-Crisioni 2017). A comparison with disaggregated travel times suggests that using population-weighted centroids provides more accurate results than geometric centroids or areal-based methods (Stepniak and Rosik (2015). Pereira et al. (2019) investigated the equity effects of transport infrastructure in Rio de Janeiro, Brazil, taking MAUP into account. The study employed three analyses that utilised two hexagonal grids and traffic zones. The findings indicate that while the direction of the relationship between accessibility increments and residents' income remained stable, there were significant differences in the magnitude and significance of this association across different spatial aggregations.

Although there are recent relevant contributions of MAUP and accessibility, there are no reports examining the interactions with refinements of accessibility measures, e.g. the consideration of matching opportunities.

2.2 Equity evaluation of transport in transport infrastructure

Transport equity concerns the adequate allocation of transport resources among different population groups and places (Lucas et al. 2019). This issue can be framed within the context of social justice, which is often defined as the “morally proper distribution of benefits and burdens among members of a society” (Lucas et al. 2019, 7). This definition captures the essence of transport equity by incorporating three key elements (ibid): 1) the resource being distributed, such as mobility or accessibility; 2) the social attributes of society's members; and 3) the moral principles used to assess whether a given allocation is fair or equitable (Martens and Bastiaanssen 2019).

The ‘equity standard’, implicit in the third point, is major topic of debate in literature. The typical questions involve, for example, should everyone receive the same amount of the good or carry the same burden, should populations in most need receive the most, etc. Karner et al. (2024) discuss two main moral philosophies addressing these questions, namely the egalitarian and the sufficientarian. The former addresses the systematic

exclusion or reduction of opportunities for neglected populations. The latter is about ensuring that minimal resources are allocated in a way that allows people to meet essential needs.

Location-based accessibility measures have been used as a key benefit being distributed in the evaluate the performance of public transport infrastructure in policy reports and research (Ermagun 2021; Liu, Porr, and Miller 2022; Conway, Byrd, and Van Eggermond 2018; Malekzadeh and Chung 2020; Palmateer et al. 2021). One motivation is because the use of realistic travel times can reflect important elements of a public transport system, such as the topology of the network, spatial network, and operation characteristics.

In line with its philosophical and moral perspective, the operationalisation of the accessibility poverty notion requires the definition of a sufficiency threshold. Accessibility poverty draws inspiration from income poverty literature and adopts an analogous poverty line (Martens, Bastiaanssen, and Lucas 2019). For analytical purposes, studies often utilise a percentile of the number of accessible opportunities. For example, Karner, Pereira, and Farber (2024) employs the population-weighted median of job accessibility in the Washington metropolitan area. Similarly, Martens, Bastiaanssen, and Lucas (2019) introduces a combined index that incorporates the percentile of employment accessible by car alongside income deprivation, indicating that an individual or group is considered to be in accessibility poverty if they face both low public transport accessibility and income deprivation. This approach is founded on the understanding that marginalised populations rely on public transport, and low-income households may either be unable to afford the costs associated with maintaining a car or find it undesirable to do so.

Despite the growing relevance of accessibility measures in the evaluation of public transport infrastructure, knowledge of the implications of overlooking matching employment are limited in literature. In this study, we aim to examine the implications of the use of a matching accessibility measure accounting for the MAUP, and its role in transport infrastructure equity evaluation. We cover 10-year period on a empirical study which includes a range of transport projects.

3 Methods

3.1 Accessibility measures examined

For this study, a general accessibility measure, also referred to as ‘All’, is computed with the following equation:

$$A_{jt}^{All} = \sum_{k=1}^J E_k f(c_{jkt}) \quad (3)$$

A_{jt}^{All} denotes the general accessibility to employment from origin j at time t . E_k are the employment opportunities at potential destinations, and c_{jkt}^{PT} is the travel time from j to k at temporal stage t .

For this work, the implementation of the matching accessibility measure is adapted from literature according to the data available for the area of study as following (Pereira 2019; Pan, Jin, and Liu 2020):

$$A_{jtm}^{Matching} = \sum_{k=1}^J E_{km} f(c_{jkt}), \quad (4)$$

Here, $A_{jtm}^{Matching}$ is the matching accessibility measure from j , at time t , for opportunities of type m . m at j is categorised in education quintiles based on the median number of years in formal education. m at k is based on the average salary paid at destinations classified by quintiles as well. Only employment opportunities E matching the same education quantile between j and k are considered in this measure (when $j \neq k$). This reflects that residents would deem valuable only employment opportunities that match their education level. In the area of study, this is supported by the positive relationship found between education level and wages in empirical studies (Nuñez, Paredes, and Garduño-Rivera 2017; Rojas, Angulo, and Velázquez 2000). An exception occurs when $j = k$. In this case, employment opportunities in the origin are considered as available. This is to avoid the assumption implying that some zones have accessibility values equal to 0. The specific data sources and procedures are detailed in the next sub-sections.

3.2 Area of study

Greater Mexico City (GMC) (*Zona Metropolitana del Valle de México*) is located in the south-centre of Mexico. In 2015, GMC had nearly 21 million inhabitants (INEGI 2015b). The official delimitation consists of 76 adjacent municipalities, which are part of three different states, namely: (1) *Ciudad de México* (16 municipalities); (2) *Hidalgo* (1 municipality), and (3) *México* (59 municipalities), as shown in Figure 1. The total GMC area is 7,889 km², and it is home to 17% of the national population.

3.2.1 The main public transport network (MPTN)

The focus of this work is on *rapid transit* and *semi-rapid transit* infrastructure, excluding specialized modes i.e. *aerial tramway*. The combination of public transport modes included in the analyses is referred to as the *main public transport network* (MPTN). Figure 1 shows the MPTN in GMC in the year 2019. This consists of a variety of modes operated by the following main agencies: The metro system, directly operated by the state-owned company *Sistema de Transporte Colectivo* (STC) and designated as Metro; two BRT systems, namely *Metrobus* (MB) in Ciudad de México and Mexibus (MXB) in the state of México; One LRT line operated by STE; and one regional rail line designated as *Sub*.

The period of study was divided into seven temporal stages to reflect significant changes in the MPTN over time. Initially, changes were identified by the physical alterations in the MPTN, such as the opening of a new corridor or the extension of an existing one. Table 1 details these changes and their corresponding temporal stages from 2010 to 2019. The stages were grouped based on the time interval between changes,

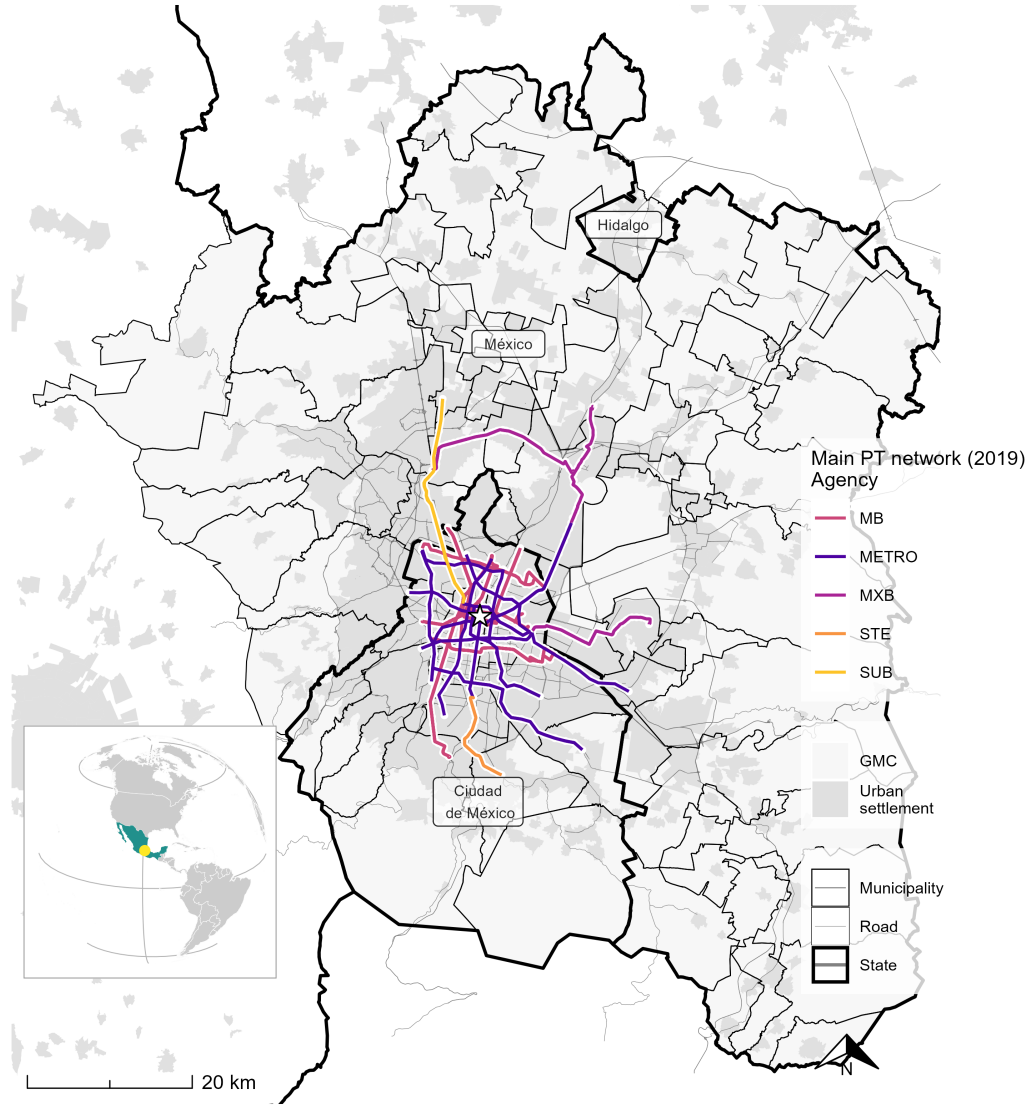


Figure 1: Main public transport network (MPTN). Greater Mexico City, 2019.

Source: Verduzco-Torres (2023) based on Marco Geoestadístico Nacional 2014 Versión 6.2 (INEGI, 2014); Delimitación de las zonas metropolitanas México 2015 (INEGI, 2018); Secretaría de Movilidad de la Ciudad de México (SEMOVI, n.d.); Sistema de Transporte Masivo y Teleférico del Estado de México (SITRAMyTEM, n.d.); OpenStreetMap (OSM, n.d.).

Table 1: Main public transport network extensions between 2010 and 2019 in Greater Mexico City grouped by temporal stage.

Corridor	Agency	From station	To station	Opening date	Previous change*	GTFS source
Stage 7: 2018 to 2019						
7	MB	Indio Verdes	Campo Marte	2018-03-01	26	2019-01-09
Stage 6: 2016 to 2018						
6	MB	El Rosario	Martín Carrera	2016-01-01	12	2017-09-02
Stage 5: 2015 to 2016						
2	MXB	La Quebrada	Las Americas	2015-01-01	14	2015-01-28
Stage 4: 2013 to 2015						
5	MB	Río de los Remedios	San Lázaro	2013-11-01	6	2014-12-30
3	MXB	Chimalhuacán	Pantitlán	2013-05-01	7	2014-12-30
Stage 3: 2012 to 2013						
12	Metro	Mixcoac	Tlahuac	2012-10-01	6	2013-10-21
4	MB	Buenavista	San Lazaro Aeropuerto	2012-04-01	14	2013-10-21
Stage 2: 2010 to 2012						
3	MB	Tenacayuca	Etiopia	2011-02-01	4	2013-10-21
1	MXB	Ciudad Azteca	Ojo de Agua	2010-10-01	10	2013-10-21
Stage 1: Before 2010						
2	MB	Tepalcates	Tacubaya	2009-12-01	NA	2013-10-21

Note:

*Months relative to the previous modification of the main public transport network.

with modifications occurring within six months being grouped together in the same stage. This criteria follows previous empirical studies and assumes that accessibility benefits are perceived as simultaneous when changes happen within short timeframes (Ahlfeldt 2013; Gibbons and Machin 2005).

The MPTN stages and corresponding corresponding corridors are in Figure 2.

3.2.2 Spatial analytical framework

Various spatial analytical schemes (SAS) are considered to address the research questions related to the MAUP. The area of study is systematically partitioned in uniform cells, j , of various gradual sizes to examine the scale effects, drawing on Wong, Lasus, and Falk (1999). The criteria and implementation employed is adopted from empirical accessibility studies, using hexagonal cells of 0.5 km, 1 km, 2 km, and 4 km (Pereira 2018, 2019). This is to (i) maintain consistency in this research area, (ii) allow comparability, (iii) exploit existing analytical tools and resources and, (iv) augment previous findings in a different context. A further scheme defined by the official post code zones was considered to account for zoning.

Table 2 summarises the main characteristics of each SAS for cells containing residents or economic activity. The total number of cells ranges from 13,300 in the highest resolution grid (0.5 km) to 410 in the lowest (4

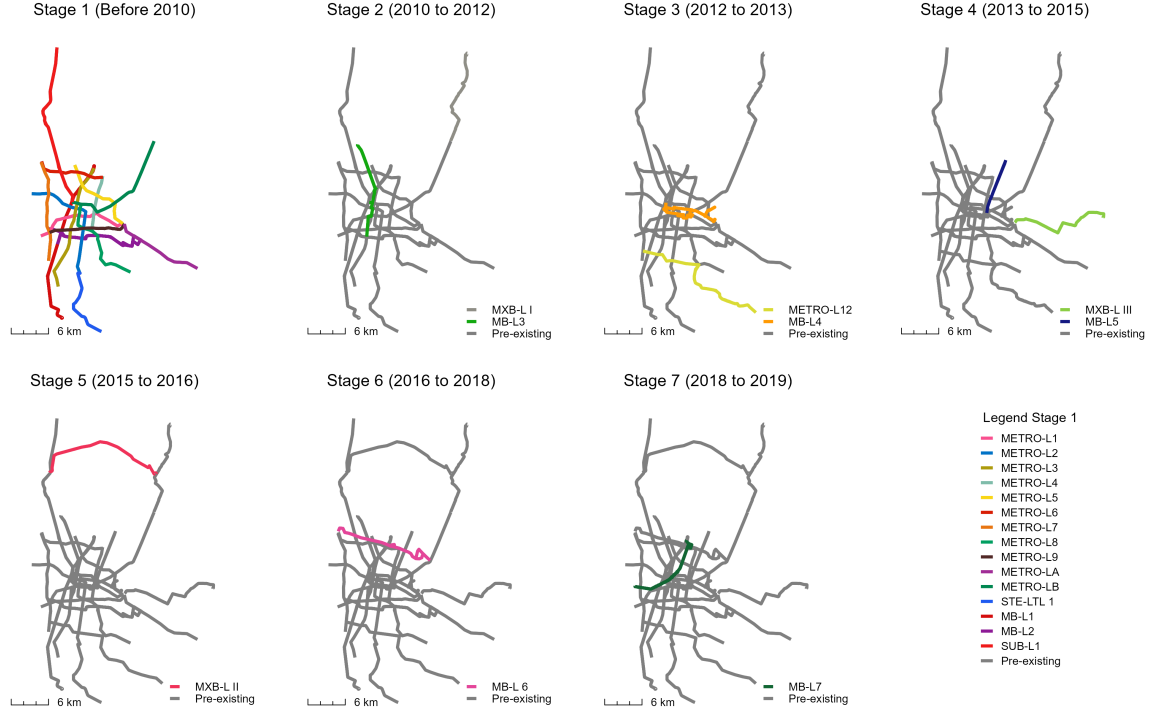


Figure 2: Temporal stages of the main public transport network. Greater Mexico City, 2010 to 2019.
Source: The author based on Semovi (n.d.) and SITRAMyTEM (n.d.).

Table 2: Spatial analytical schemes (SAS) overview.

	Grid 0.5 km	Grid 1 km	Grid 2 km	Grid 4 km	Post code
Count	13 297	3 972	1 264	413	2 377
Mean area (Sq. km)	0.22	0.86	3.44	13.55	1.94
Mean population (Inhabitants)	1 498.13	5 093.21	16 324.62	50 187.76	8 317.75

Source:

Censo de Población y Vivienda 2010 (INEGI, n.d.)

Sistema para la Consulta de Información Censal 2010 (INEGI, n.d.)

Mapa de Ubicación de Códigos Postales (V5). Correos de México, 2019

km), with surface areas varying between 0.22 km^2 and 13.5 km^2 . On average, the smallest cells contain only 1,500 inhabitants, while the largest have over 50,000.

The polygons were created manually in R programming language following previous work (Pereira 2018). Post code polygons are sourced from the Mexican Post Office (*Correos de México*) ‘Version 5’.

3.3 Data

3.3.1 Demographic

The demographic comes from the 2010 Population Census (INEGI 2012). These data were selected for their detailed disaggregation at the city block level, including key fields such as total population and average years of education completed by individuals aged 15 and older.

The count data included were transferred to the SAS using the *areal weighting method* (Saporito et al. 2007). This reassigns the data according to the proportional overlay of the source and target areas. The average education level was assigned using the *population weighting method*. In addition, this takes into consideration the number of inhabitants and their type.

3.3.2 Employment

Employment data are sourced from the 2014 Economic Census (*Censos Económicos 2014*) (INEGI 2015b) and accessed at the city block level via the Microdata Lab and Remote Processing service (<https://www.inegi.org.mx/microdatos/>). The dataset includes: 1. Number of establishments (fixed locations producing goods or services); 2. Employed personnel (individuals active in establishments, including volunteers and family members), referred to as ‘employment’; and 3. Total gross remuneration paid for the relevant year.

Employment information which came aggregated at higher geographic level than the city-block due to privacy (<8%) were re-allocated using proportional estimates based on the spatial distribution from the 2015 DENU (INEGI 2015a). This is an open-access, point-level dataset of individual establishments. The employment data was assigned from city blocks to SAS following the areal weighting method too (Saporito et al. 2007).

The average income was computed for each areal unit based on the total number of employment positions and the total gross remuneration paid.

3.3.3 Public transport timetables

The estimated travel time used in accessibility measures is based on scheduled public transport services. The Transport Authority in Mexico City (*Secretaría de Movilidad de la Ciudad de México*, Semovi) released the first *general transit feed specification* (GTFS) files in 2013, which were accessed via the Open Mobility website (<https://openmobilitydata.org/>). These datasets cover the agencies operating most MPTN services in GMC. However, there are some limitations: firstly, the geographic coverage primarily includes services within Ciudad de México, excluding BRT services provided by MXB. Secondly, the temporal coverage starts in 2013, with updates occurring only sporadically, approximately once a year.

To overcome the first limitation, the geographic coverage of the official GTFS data was augmented to include three BRT corridors operated by MXB. This was carried out *manually* based on unstructured information

provided by the Transport Agency of the State of Mexico (*Sistema de Transporte Masivo y Teleferico del Estado de México*, Sitramytem). This corresponded to services for 2019 only.

Regarding the second limitation, the MPTN’s temporal stages were reconstructed using the GTFS standard. The primary input for each stage was the closest available official GTFS data reflecting changes in the MPTN, with only the relevant services for each corridor in their respective stages being retained. For MXB services, 2019 data was considered due to the lack of earlier details, assuming uniform characteristics of the BRT corridors across the stages. Similarly, for other corridors included in early stages, services were assumed to reflect 2013 levels due to the unavailability of earlier data. All non-MPTN services were excluded, thereby focusing exclusively on the benefits of the MPTN.

3.3.4 Road network and pedestrian infrastructure

The road and pedestrian infrastructure network is sourced from *OpenStreetMap* (OSM) (Manually downloaded on 30th October 2019 through a web export tool available in <https://export.hotosm.org/es/v3/>). This enables multi-modal journeys to realistically represent the access/egress stage to a public transport service when combining walking and public transport.

3.4 MPTN travel times and impedance parameters

3.4.1 Modeled travel time

Travel time matrices were calculated for each origin-destination (OD) pair across each SAS using a combination of walking and public transport within the main public transport network. OD points j and k , respectively, are represented by population-weighted centroids of spatial units containing economic activity (from the 2014 Economic Census) or demographic data (from the 2010 Population Census). Travel times are capped at 180 minutes, based on patterns observed in the 2017 Travel Survey.

Travel times were modelled using a door-to-door approach with the **r5r** software package for the R programming language (Saraiva et al. 2021; R Core Team 2021). This approach considers access, waiting, onboard, transfer, and egress times. All-to-all travel time matrices were generated for each SAS across seven temporal stages using reconstructed GTFS files and road/pedestrian networks. The estimated travel time considers the median time within a three-hour time window between 8 a.m. and 11 a.m. The maximum walking distance to access and egress public transport or in walking journeys only is limited to 40 minutes, approximately 3.2 km at an average walking speed of 4.8 km/h.

The internal-zone travel-times were estimated based on the surface area as following: $d_{j=k} = \frac{1}{2} \sqrt{\frac{area_j}{\pi}}$ (Frost and Spence 1995; Stepniak and Rosik 2015). The metric distance was transformed to travel time using the average speed modelled for short journeys (less than 3 km).

3.4.2 Spatial decaying function and calibration of accessibility parameters

The spatial decaying function and parameters for accessibility measures were determined using estimates from a trip distribution model, following Handy and Niemeier (1997). This analysis use data from the 2017 Travel Survey (INEGI 2018), focusing on a weekday and journeys including public transport or walking.

Alternative specifications were estimated within a doubly constrained gravity model framework, expressed as following:

$$T_{jk} = A_j O_j B_k D_k f(c_{jk}) \quad (5)$$

where

$$A_j = \left[\sum B_k D_k f(c_{jk}) \right]^{-1} \quad (6)$$

and

$$B_k = \left[\sum A_j O_j f(c_{jk}) \right]^{-1} \quad (7)$$

with balancing factors A_j and B_k calculated iteratively. T_{jk} are commuter flows between origins (j) and destinations (k), with constraints on the number of journeys at each, i.e. O_j and D_k . $f(c_{jk})$ is a spatial deterrence or decaying function. The inverse power ($c_{jk}^{-\beta}$) and negative exponential ($e^{-\beta c_{jk}}$) decaying functions were tested. β represents the deterrence parameter. c_{jk} is the observed travel time.

A modified specification aimed at capturing an add-up cost for inter-zone journeys or analogously a benefit of residing and working within the same zone is also considered (Thorsen and Gitlesen 1998). This includes a parameter μ which affects only the diagonal elements of the OD matrix. The term is given by the following expression:

$$\mu \delta_{jk} \quad (8)$$

where Kronecker delta takes the following value:

$$\delta_{ij} \begin{cases} 0 & \text{if } j \neq k \\ 1 & \text{if } j = k \end{cases} \quad (9)$$

The simultaneous calibration of the model parameters β and μ is achieved via optimization by minimizing the standardized root mean square error (SRMSE) using the *Nelder-Mead's simplex algorithm* as implemented in base R (R Core Team 2021).

The preliminary analyses shows that the negative exponential function provided more accurate estimates compared to the inverse power. Also, the use of the within-zone parameter μ improves the estimates.

Therefore, the selected decaying function used in the accessibility measures in Equation 3 and Equation 4 is the modified negative exponential including μ , as shown below:

$$f(c_{jkt}) = e^{(-\beta c_{jkt}^{PT} + \mu \delta_{jk})} \quad (10)$$

where, c_{jkt}^{PT} is the modelled travel time from j to k at the MPTN temporal stage t . The optimized parameters used are $\beta = 0.031$ and $\mu = 1.662$.

3.5 Analytical framework

3.5.1 Accessibility indicators

To focus on the key spatial differences between generic and matching accessibility measures and the role of the MAUP, the first section of the analysis summarises the level of MPTN accessibility over the period of study as following:

$$A_j^{All} = \text{median}(A_{jt}^{All}) \text{ for } t = 1, 2, \dots, 7, \quad (11)$$

and

$$A_{jm}^{Matching} = \text{median}(A_{jtm}^{Matching}) \text{ for } t = 1, 2, \dots, 7. \quad (12)$$

A similarity rate is computed to compare the All and Matching measures as following:

$$\text{Similarity rate} = \frac{A_{jm}^{Matching}}{A_j^{All}}. \quad (13)$$

This ranges between 0 and 1, where values near to 0 indicate low similarity and 1 a high similarity rate.

3.5.2 Accessibility equity measures

For the transport equity evaluation of infrastructure projects, the prevalence of accessibility shortfalls measure is adopted from Martens, Bastiaanssen, and Lucas (2019). This is expressed by the equation below:

$$H_t = \sum_{j=1}^P I(A_{jt} < z) \quad (14)$$

Where, H_t represents the number of people whose accessibility level falls below the established accessibility sufficiency threshold at temporal stage t , P is the total population, A_{jt} is the accessibility level of at location j and temporal stage t , z is the sufficiency threshold, and $I(A_{jt} < z)$ is an indicator function that equals 1 if

y_{it} is less than the sufficiency threshold (z), and 0 otherwise. We also use the headcount ratio $P_0 = H_t/N$, where a larger share indicates a less fair transport system. Given that this is a standardised measure, we can compare the different temporal stages.

We use a range of sufficiency thresholds z using the 10, 15, 20, and 30 percentile of accessibility by car for each corresponding measure, i.e. All or Matching. Accessibility by car is estimated using Equation 3 and Equation 4 where $f(c_{jk}) = \exp(-\beta c_{jk}^{Car})$. c_{jk}^{Car} is the estimated travel time by car. This approach draws on Martens, Bastiaanssen, and Lucas (2019).

3.5.3 Indicators of spatial association

Two spatial statistics techniques are used to measure the spatial difference between the All and Matching measures. The first, the Global Moran's I statistic, which measures spatial autocorrelation, is expressed as (Moraga 2023):

$$I = \frac{N \sum_j \sum_k w_{jk} (x_j - \bar{x})(x_k - \bar{x})}{\left(\sum_{j \neq k} w_{jk} \right) \sum_j (x_j - \bar{x})^2}$$

Where, N is the number of individual units in each SAS, x_j and x_k are the accessibility levels at location j and k , respectively, $\bar{x}$ is the mean accessibility, w_{jk} represents a binary spatial weight between locations i and j , and the term $\sum_{j=1}^N \sum_{k=1}^N$ is the sum of the spatial weights. w_{jk} is an element of the spatial weight matrix W , which was computed based on 6-nearest neighbours of the OD points with the `spdep` package (Bivand 2021).

Local Moran's I is used to account for local indicators of spatial association (LISA), defined as:

$$I_j = \frac{N(x_j - \bar{x})}{\sum_k (x_k - \bar{x})^2} \sum_k w_{jk} (x_k - \bar{x})$$

A high value for I_j suggests that the area is surrounded by areas with similar values, indicating that it is part of a cluster of high observations, low observations, or moderate observations. Conversely, a low value implies that the area is surrounded by dissimilar values, marking it as an outlier and indicating that the observation of area j differs from most or all of its neighbours.

We compute a map of p-values that represent the probability of exceeding the observed values under the null hypothesis, allowing for the identification of significant clusters.

Table 3: Summary statistics of accessibility measures over various SAS.

SAS	All	Matching	N
Grid 0.5 km	59.4 (121.3) [0.9, 50.8]	25.5 (71.4) [0.3, 9.2]	12 535
Grid 1 km	55.1 (120.8) [0.8, 31.7]	28.5 (83.1) [0.3, 10.8]	3 765
Grid 2 km	54.1 (130.4) [0.8, 25.0]	37.1 (108.6) [0.5, 16.6]	1 182
Grid 4 km	62.8 (166.6) [0.9, 29.2]	54.9 (158.1) [0.8, 27.5]	387
Post code	107.1 (149.2) [4.8, 157.3]	41.5 (85.7) [2.1, 33.5]	2 301

Note:

Descriptive statistics: Mean (SD) [1st-Q, 3rd-Q]. Figures shown in thousands.

4 Findings

4.1 Spatial analysis

Table 2 summarises descriptive statistics for both accessibility measures across various SAS. The mean values for the measure considering all type of employment range from 59,400 in the 0.5 km grid to 107,000 in the postcode schemes, while the matching measure ranges from 25,500 to 41,600. This indicates a smaller scale for the latter due to restrictions imposed in employment opportunities. Notably, the mean of the matching measure increases steadily with scale, from 25,500 in the 0.5 km grid to 54,900 in the 4 km grid, whereas the ‘All’ measure shows no consistent trend. The postcodes scheme results in the largest mean values. This can be explained by zoning effects. Specifically, peripheral regions, which are associated with low accessibility, tend to be larger, thereby reducing the number of low-accessibility value units. Meanwhile, equal-sized grids fragment these regions of the city into a larger number of units, thereby pulling the mean lower.

The series of maps in Figure 3 show the modelled MPTN accessibility to employment aggregated at various SASs (columns) by type of accessibility specification (rows). The accessibility values are presented in decile breaks. The top row illustrates accessibility to employment considering all opportunities. High resolution grids, such as 0.5 km or 1 km, highlight the relevance of both central areas and radial public transport corridors, resembling a star-shaped structure. The post code SAS partially reflects some of these patterns too. Examples of corridors connecting non-central areas include the regional rail Line 1 *Buenavista-Cuautitlán* (SUB-L1) to the northwest and the Metro line B (METRO-LB) combined with the Mexibús BRT Line I (MXB-LI) to the northeast. The SUB-L1 generates high-accessibility “islands” around stations, even far from the CBD. In contrast, the northeast corridor’s effects are most pronounced along the METRO-LB segment, diminishing further from the CBD and in areas served by the BRT MXB-L1. These varying patterns arise mainly from the physical and operational characteristics of the transport modes and land use, i.e. operational speed, distance between stations, and proximity to main employment centres.

Median MPTN accessibility to employment by public transport. Greater Mexico City.
Accessibility is computed as the median of the period of study for each spatial unit.

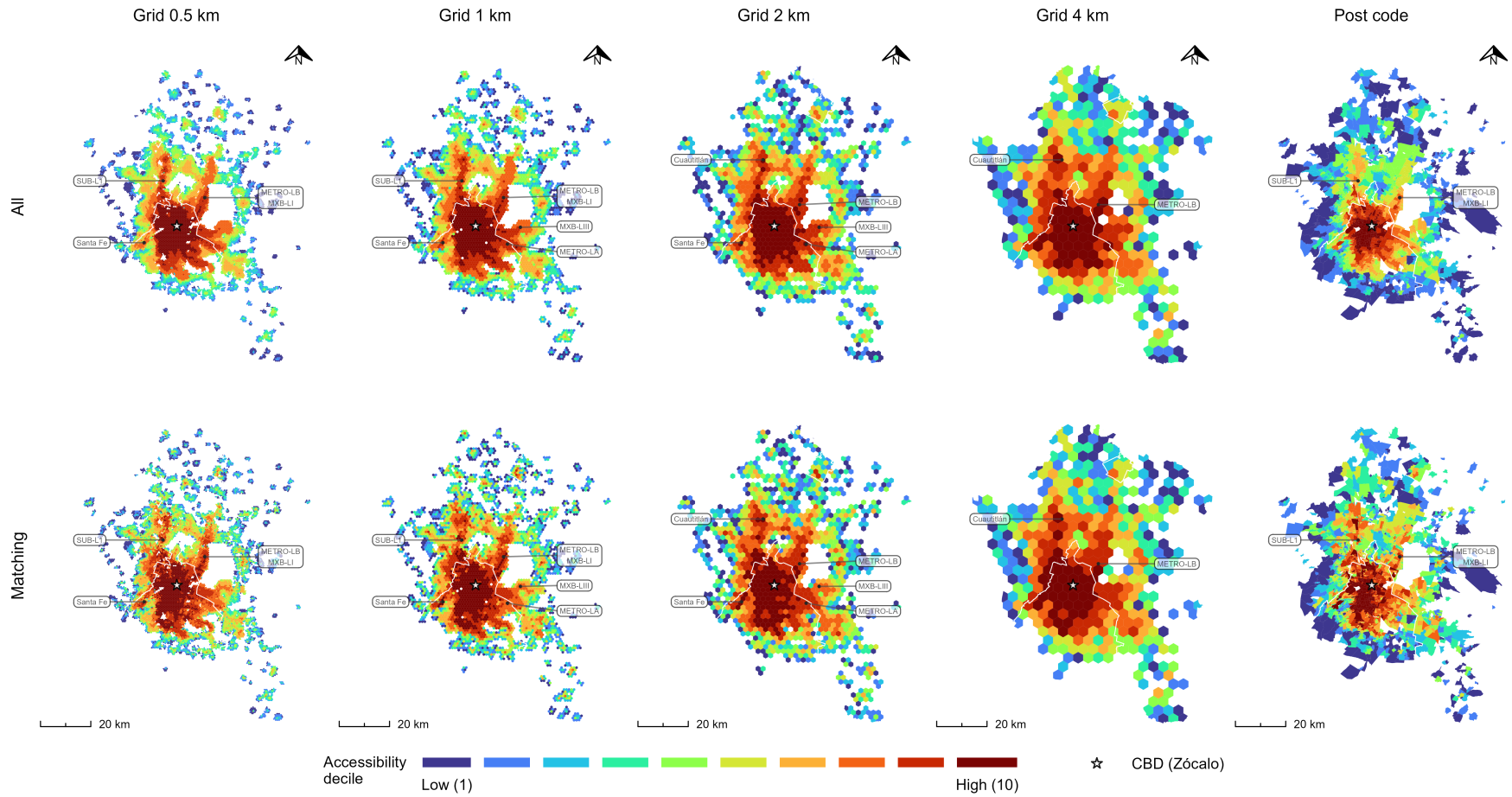


Figure 3: Median MPTN general and matching accessibility over various spatial analytical schemes. Greater Mexico City.
Source: The author based on Semovi (n.d.) and SITRAMyTEM (n.d.).

Table 4: Correlation coefficient over various SAS

Type	SAS				
	Grid 0.5 km	Grid 1 km	Grid 2 km	Grid 4 km	Post code
<i>Bivariate correlation</i>					
Pearson	0.860***	0.879***	0.937***	0.989***	0.780***
<i>Global spatial autocorrelation</i>					
All	0.926***	0.888***	0.812***	0.690***	0.929***
Matching	0.845***	0.795***	0.742***	0.657***	0.737***

Signif. Codes: ***: >0.01 , **: >0.05 , *: >0.1

SAS = Spatial analytical scheme.

Whilst high-resolution SASs reflect detailed characteristics of the MPTN, many of these features are smoothed into polygons or zones in low-resolution schemes, e.g. 2 km or 4 km grid. At this scales, the star-shaped structure develops into concentric-like bands or accessibility rings. At the same time, the differences between the measure are less visible as the scale increases.

The results of a bivariate Pearson correlation test between the generic measure and the matching are presented in Table 4. It is shown that the correlation coefficient increases from 0.86 in the 0.5 km scheme to 0.99 in 4 km one. This shows that the added information by the refined matching measure is gradually lost as the scale of spatial units increases. This is likely because accessibility measures emphasize employment that are in proximity. Therefore, large units mostly capture the within-zone opportunities. The post code schemes show the lowest similarity.

Another general observation from Figure 3 is that the matching measure produces more spatially heterogeneous patterns than the measure considering all types of employment. This is confirmed by the global spatial autocorrelation test results in Table 4. Moran’s I statistic ranges from 0 (no spatial autocorrelation) to 1 (high spatial autocorrelation). The results show that the matching measure consistently produces lower values than the generic accessibility measure. However, the gap narrows as the scale increases. For example, the difference is 0.08 points in the smallest grid but only 0.03 points in the 4 km grid. Postcode boundaries, however, show the largest gap, nearly 0.2 points.

4.1.1 General versus matching accessibility and equity implications

In the next step of the analysis, the accessibility measures are directly compared using a similarity rate, where values close to 0 indicate low similarity and values close to 1 indicate high similarity. We are also interested in whether the differences between matching and generic accessibility measures follow a spatial pattern. Figure 4 displays the local Moran’s I clustering results for the similarity rate across each SAS. High- and mid-resolution SAS reveal a consistent West-East divide, with High-High clusters located towards the West and Low-Low clusters towards the East, using the CBD as a reference point. This suggests that differences between the accessibility measures tend to be stronger in the eastern areas of the city.

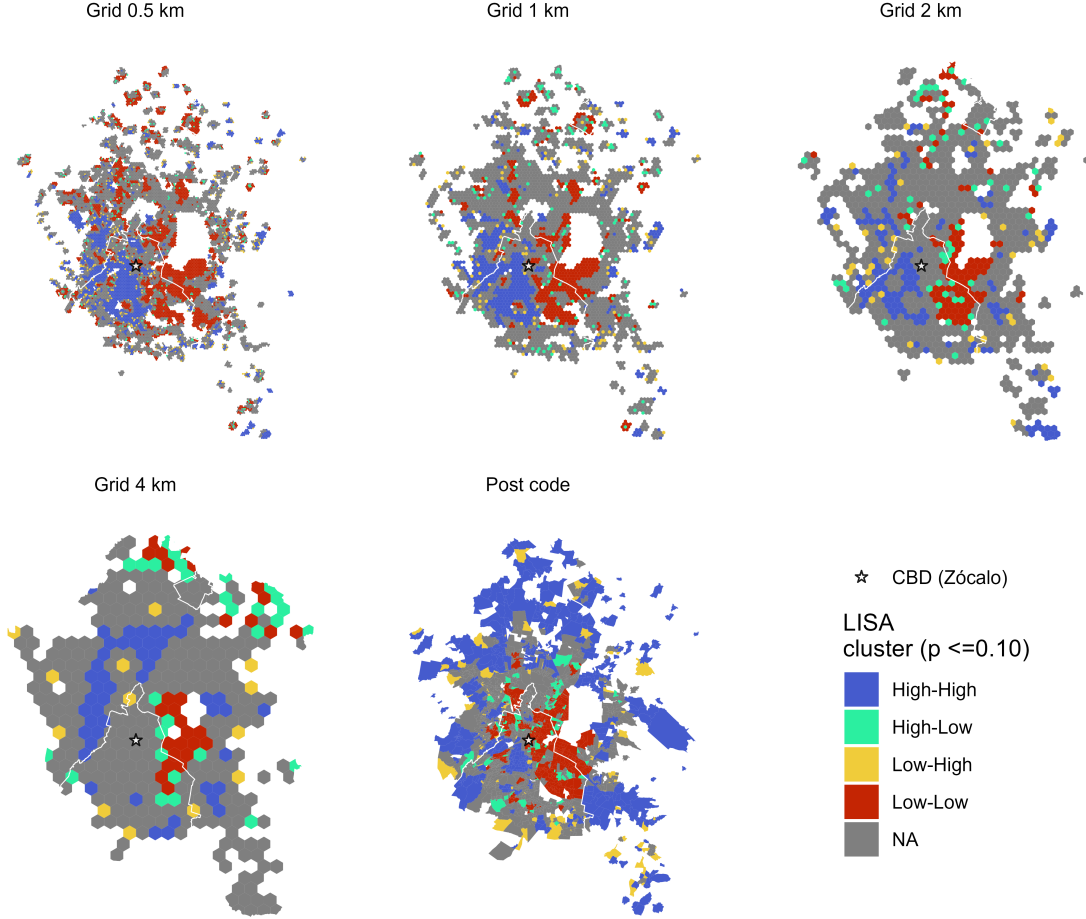


Figure 4: Local indicator of spatial association on similarity rate between generic and matching MPTN accessibility over various SAS.

SAS = Spatial analytical scheme; MPTN = Main public transport network.

While the postcode and 4 km grid show a similar trend for Low-Low clusters, the High-High patterns are less consistent across SAS. In particular, the postcode SAS show substantial differences in the spatial distribution of High-High clusters, with many located in peripheral areas. This can be explained because postcode zones in the periphery tend to be larger. Thus, travel times to jobs outside the zone are generally long and the level of accessibility will likely focus on within-zone opportunities. Since the matching measure accounts for all within-zone employment opportunities, the similarity rate is high in many peripheral zones. This is one way in which zoning can interact in the differences between measures.

The overall West-East divide revealed in the LISA analysis is generally consistent with well-known spatial trends of key socioeconomic indicators in GMC, i.e. with socio-economically advantaged population residing in the West (Ortega-Avila 2023). To examine potential systematic differences, the LISA clusters are examined by their social composition. Figure 5 displays the proportion of the population which were classed in High-High or Low-Low clusters by education quintile for all SAS. This only includes significant clusters. It is clear that

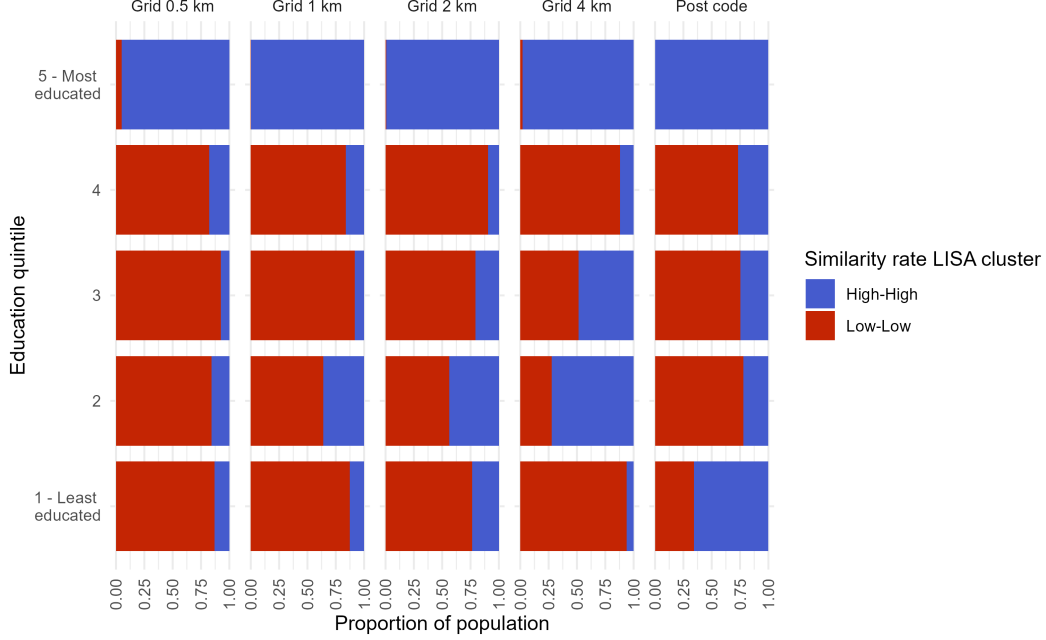


Figure 5: Composition of local indicator of spatial association clusters on similarity rate by education quintile. LISA outlier clusters are not shown.

the accessibility measures are very similar for the vast majority of the most educated population across all SAS. By contrast, most of the population in less educated groups fall within Low-Low clusters.

The post code scheme shows an exception to this trend. According to this SAS, most of the least educated population reside in units with high similarity. Likely, this is related to the characteristics presented in the previous paragraph, where periphery zones tend display high similarity rates. At the same time, these areas tend to be populated by low-educated groups, particularly to the North and North-East.

4.2 Main public transport network assessment

In this section, the focus shifts from spatial variations to temporal changes, aiming to capture the effects of public transport network extensions. To better distinguish these changes, the spatial unit of aggregation is kept fixed. Therefore, the following analysis uses only the 0.5 km grid, as it is the smallest and most detailed SAS.

Figure 6 provides an overview of the relative average changes in the network weighted by the population across various temporal stages, using the distinct accessibility measures. The baseline is set to the ‘Before 2010’ period. Generally, both measures show consistent trends, with notable increases in accessibility during the Stage 3 and Stage 7. These include the introduction of a 25-kilometre metro line (METRO-12) connecting south-east zones to the MPTN, and a BRT line (MB-L7) in a central area of the MPTN, respectively. Extensions in other stages show minor average increases.

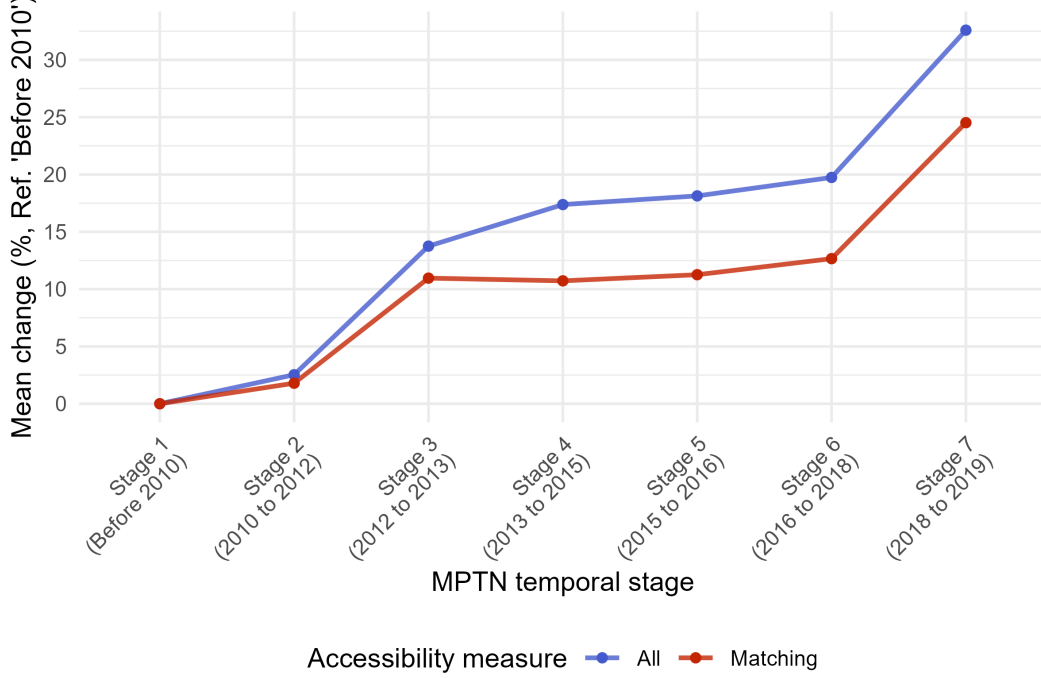


Figure 6: Relative change of accessibility population weighted mean over various temporal stages.
 <!-- ... -->

Interestingly, the general accessibility measure in Figure 6 shows moderate increases in Stage 4 (2013-2015) and Stage 5 (2015-2016), while the matching measure records little to no change, and even a slight decline during these stages. This divergence suggests that while the infrastructure improvements introduced in these periods allowed access to additional employment from certain residential zones, the jobs at destinations did not match the educational qualifications in the areas of influence.

We also draw on the results from a transport equity perspective. An advantage is that it can distinguish between populations that were already served at an ‘adequate’ level before an intervention and those who were not. Figure 7 shows the head count ratio which are under various minimal accessibility thresholds over the MPTN temporal stages for the ‘All’ and ‘Matching’ measures. From the plot, a general reduction in the population under the accessibility thresholds can be observed. This is in the direction expected.

There are additional insights to draw from the results. First, the scale of the ratio is smaller for the matching measure, with the gap narrowing as the percentile increases. This relates to the different density distributions of the measures, where the ‘Matching’ measure is more skewed than the ‘All’ measure. Still, both measures show parallel trends for most stages, with a few exceptions. For example, the ‘All’ measure consistently declines between Stage 1 (‘Before 2010’) and Stage 4 (‘2013 to 2015’) across all thresholds, while the ‘Matching’ measure only begins to reflect this trend after the 15th percentile, showing similar trends by the 30th percentile.

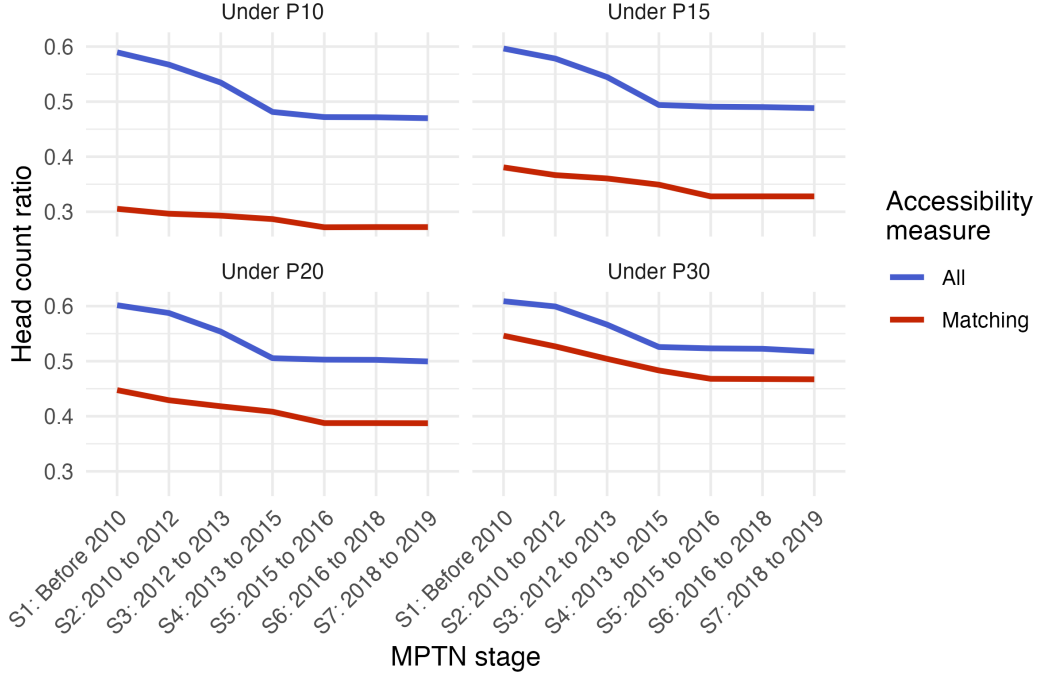


Figure 7: Equity analysis over various temporal stages. S = Stage.
 <!-- ... -->

Another distinction is that the ‘Matching’ measure shows significant improvement in Stage 5 (‘2015 to 2016’) for all thresholds, which is not captured by the ‘All’ measure. This improvement corresponds to the BRT corridor MXB-LII, located in the periphery of the MPTN and connecting key corridors in the northern areas of the city. This suggests that some northern populations were previously considered well-served by the generic measure, even though the employment opportunities did not match their qualifications. This means that the opening of MXB-LII enabled access to matching employment opportunities for those who already had access to a minimum number of general opportunities.

The global population-weighted average in Figure 6 highlights large accessibility benefits introduced in Stage 7 (2018-2019). However, the equity analysis results in Figure 7 show consistent negligible impacts across all percentiles and measures in this stage. This is because the MB-L7, implemented during this stage, is centrally located within the MPTN and primarily serves to articulate existing network corridors. As a result, the additional accessibility to employment was concentrated in areas that already exceeded the examined thresholds.

Figure 8 shows the net population taken out from various accessibility thresholds by education quintile across the MPTN stages. Between Stage 2 and Stage 4, the groups benefiting from the infrastructure are concentrated in the intermediate quantiles, a trend more pronounced in the general accessibility measure. The matching measure remains consistent for some stages, particularly for the 30th percentile threshold. Interestingly, the matching measure reveals a progressive distributional effect in Stage 5, where the benefits go

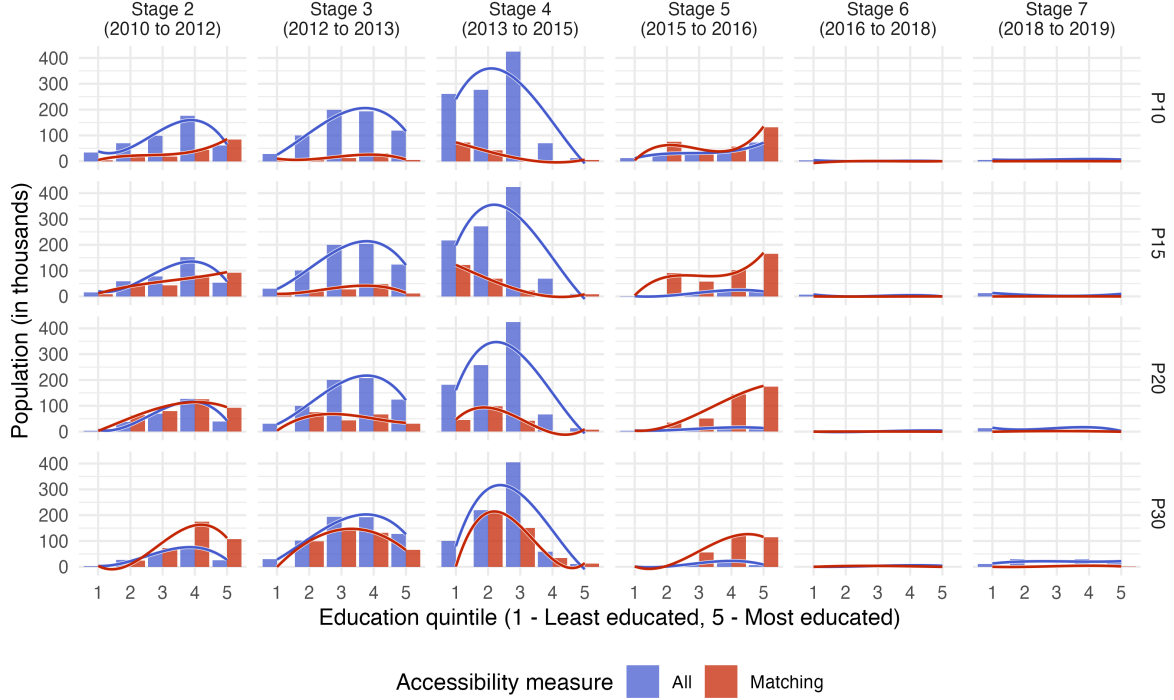


Figure 8: Equity analysis by education level over various temporal stages.

<!-- ... -->

to lower-educated groups, between the first and third quintile. This makes sense, since the MXB-LIII enabled direct connection to the MPTN to two densely populated municipalities characterised by high-deprivation levels, i.e. *Ciudad Nezahualcóyotl* and *Chimalhuacan*. However, these effects diminish at higher accessibility thresholds, likely because low-educated populations reside in less central areas. Thus, as the threshold increases, lower-educated zones fall short, while areas with higher education levels are considered beneficiaries.

Figure 8 also highlights the matching employment benefits in Stage 5 (2015-2016), as discussed earlier. Interestingly, these benefits primarily went to the more educated population between the third and fifth quintiles, a pattern consistent across all thresholds. As noted, no significant population was taken out of the minimal accessibility thresholds in Stages 6 and 7.

5 Discussion

The results examining the spatial differences between generic and matching accessibility measures showed important interactions with the MAUP. Specifically, the larger the scale, the less relevant the distinction of matching accessibility becomes, as indicated by the increased correlation. To some extent, this is explained by the tendency of larger units to emphasise within-zone opportunities, given the increased travel costs for residents to leave the unit or reach another one. Thus, at larger scales, accessibility measures tend to reflect employment density. Interestingly, the postcode retained matching employment information to a very good

extent, as evidenced by the low correlation. By implication, and as shown visually, the role of public transport infrastructure gradually diminishes with the use of larger scales. Similarly, postcode units demonstrated a retention of some of these features. Analysts who aim to account for matching characteristics and the role of transport infrastructure should take these effects into consideration.

Furthermore, our results indicate that zones with significant similarity between the ‘All’ and ‘Matching’ measures were predominantly composed of the most educated population. This reflects a West-East divide, aligning with the well-documented spatial inequalities in GMC, such as those related to household income and car ownership (Guerra 2015), healthy food environments (Ortega-Avila 2023), and social mix (Letouzé et al. 2022). From an urban economic perspective, highly educated populations are likely to be in a position to afford residential locations that maximise accessibility to employment opportunities they consider valuable, outbidding less affluent groups (Duranton and Puga 2015). An implication for accessibility analyses that draw on the assumption of equivalent attractiveness of employment is that they might systematically overestimate the relevant opportunities available to less educated populations. This is expected to be particularly relevant in contexts with higher economic inequalities, following urban land-rent theory.

The results regarding the evaluation of public transport infrastructure over time are varied. Firstly, it is confirmed that a simple measure alone, such as the global average of accessibility, might overlook key transport equity insights (Karner, Pereira, and Farber 2024). Specifically, while certain extensions of the MPTN reflect a substantial improvement on average, an equity analysis from a sufficiency perspective shows negligible effects. We argue that both indicators supplement each other. For instance, the results highlight a concentration of the benefits in Stage 7 among populations that were already served. This finding was consistent regardless of the accessibility measure being used.

Secondly, both accessibility measures point in similar directions with parallel trends at a higher level. However, there are particularities to note. The matching accessibility measure was more sensitive to sufficiency thresholds in the equity analysis. This may be related to the observation that the distribution of the baseline used to define the thresholds presents a more skewed distribution.

The comparison between measures also showed some complementarity. For example, the fact that the matching measure reflects that there are populations benefited by particular infrastructure interventions, while the general measure does not, implies that certain group might have received ‘general’ accessibility improvements previously and, later, specific ones related to matching employment opportunities. Both indicators are useful when considering the former measure as a form of *generalized access to urban opportunities*, as the location of employment often reflects the provision of a service (Merlin 2017; Levinson and Wu 2020). Meanwhile, a distinction of matching employment opportunities is useful when applied to commuting to work patterns or labour market inequalities, for example.

Thirdly, the particular differences arising from the inclusion of matching opportunities in equity analyses vary with a range of outcomes from case to case. It is shown that the implementation of a transport project can potentially imply benefits either for well-educated or low-educated groups only, or that measures can be consistent at times. In the area of study, this is observed to be related to the location of the transport improvement with respect to the overall transport network and the demographic characteristics of the population in proximity to the public transport services. This reinforces the important role of accessibility measures in public transport evaluation, given that they are able to capture key elements of the transport system and characteristics of the population simultaneously. Thus, it is recommended to use both measures to form comprehensive insights into the impact of transport infrastructure on a case-based basis.

5.1 Limitations and future work

The present study comes with the following limitations. Specifically, the analyses assume that all populations within zones share similar demographic characteristics, e.g. the same education levels or employment opportunities. This is a broader issue in geography discussed under the ‘Ecological Fallacy’ literature (Walker 2021). This is also related to limitations of data, given that individual-level data for both jobs and residents are restricted to maintain privacy. However, this is expected to have minor implications because the original source information is aggregated in small geographic units, the urban block. Additionally, the modelled matching access measure used assumes limited social mobility, allowing only equal education-income quintiles to be matched. Additional research in different contexts can clarify the sensitivity of this assumption. Furthermore, the analyses focus on the MPTN only, which overlooks street transit services. In the case of the study, this is largely constrained by the information available, given that comprehensive structured timetable data is not accessible.

Future work is recommended to examine the sensitivity of results when using individual-level job characteristics to establish a match between residents and opportunities. Similar analyses in a diversity of contexts with a range of social and economic disparities would help to identify consistent patterns. This could help to establish the extent of the findings in our empirical case. Furthermore, additional work should examine the assumptions of matching measures and their relevance for specific applications, e.g. in the housing market or commuting to work patterns.

6 Conclusions

The present study relaxed a common assumption of accessibility analysis that access to employment is equally attractive to all types of residents. The study examines the specific differences between a generic and matching accessibility measures taking into the account the MAUP, and its relevance for equity assessments of public transport infrastructure.

It is shown that both components of the modifiable areal unit problem, namely scale and zoning, play an important role in this distinction. Specifically, the added information regarding the matching employment is gradually smoothed with increased scale. Also, the difference between measures is sensitive to zoning as post codes, for example, can over-represent the role of large zones in local spatial analyses.

A sensitivity analysis is recommended particularly in context with high social inequality. This is because the allocation of residential land use might push low-income residents more acutely with the consequence of the level of access to relevant employment being systematically underestimated for less advantaged populations. Also, the interpretation and information provided by the distinct measures will depend on specific applications. In this study, we have shown that both measures play complementary roles.

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